HUY PHAM (AEHUS)

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RESEARCH PROFILE

AI Research Engineer at Nanyang Technological University specializing in large language models and multimodal retrieval. Previously at Zalo AI, I mainly designed and developed multimodal image retrieval solutions for production-scale systems serving millions of users. Co-author of 2 top-tier conference papers, with 4+ years of experience spanning LLM pre-training, supervised fine-tuning, data synthesis, model evaluation, and production-grade AI applications.

RESEARCH & WORK EXPERIENCE

AI Research Engineer **Aug 2025 – Present**
ATMRI – Nanyang Technological University *Singapore*

- **Multi-hop RAG:** Developing retrieval-augmented reasoning frameworks for multi-hop question answering, focusing on graph-based retrieval, evidence grounding, and agentic search trajectories.
- **ATC Assistant:** Architecting a multimodal chatbot assistant for air traffic management research, enabling retrieval across heterogeneous aviation documents, structured data, and domain-specific knowledge sources.

LLM Scientist **Jul 2023 – Jul 2025**
Zalo AI – VNG Corporation *Vietnam*

- **Multimodal Image Retrieval:** Led the development of multimodal retrieval solutions for Zalo Image Search, supporting large-scale, low-latency inference for millions of users.
- **LLM Post-training:** Designed and executed large-scale supervised fine-tuning and instruction-tuning experiments for LLMs and VLMs across Vietnamese language understanding, reasoning, and multimodal tasks.
- **Human-in-the-loop Data Pipeline:** Built human-in-the-loop annotation and feedback workflows, including annotation guidelines, quality-control protocols, and iterative data refinement for high-quality instruction-tuning datasets.
- **Model Performance:** Improved Kiki LM performance on VMLU Vi-Dialog from 44% to 84%, and develop Kiki VLM to reach 43.8%, which achieve highest score on Vi-MTVQA.
- **LLM Pre-training:** Directed data-centric pre-training research for KiLM, producing nearly 2TB of cleaned Vietnamese text for training a Vietnamese large language model from scratch.

AI Research Engineer **Jan 2022 – Jun 2023**
FPT Software AI Center – FPT Corporation *Vietnam*

- **Vietnamese Language Modeling:** Co-authored ViDeBERTa, a state-of-the-art Vietnamese encoder model accepted to EACL 2023, establishing a strong baseline for Vietnamese natural language understanding.
- **Efficient Pre-training:** Conducted research on efficient transformer pre-training and model optimization for Vietnamese NLP, improving downstream performance.

SELECTED PUBLICATIONS & OPEN-SOURCE MODELS

VMLU: A Comprehensive Benchmark for Vietnamese Language Model Understanding
ACL 2025 – Co-author

- Contributed to the design and development of a comprehensive benchmark for evaluating Vietnamese LLMs across language understanding and dialogue following ability.

ViDeBERTa: A powerful pre-trained language model for Vietnamese
EACL 2023 – Co-first author

- Developed a Vietnamese pre-trained encoder model that established a strong baseline for Vietnamese natural language understanding tasks.

VinaLLaMA & Vietcuna
Co-founder and Lead Developer

- Released early instruction-tuned LLaMA-based models for Vietnamese, supporting the development of the local open-source LLM community.

Safety-Oriented Evaluation of Language Understanding Systems for Air Traffic Control
ITSC 2026 – Co-author

- This paper studies the problem of evaluating language understanding systems in safety-critical ATC environments and proposes a risk-aware, consequence-oriented evaluation framework.

Retrieval-Guided Planning for Robust Multi-Hop Reasoning
Under review – First author

- Proposed a retrieval-guided planning framework that dynamically refines search trajectories using intermediate evidence, improving grounded multi-hop reasoning.

TECHNICAL SKILLS

Research Areas	LLM Pre-training, Post-training, Multimodal Retrieval, RAG, Multi-hop QA, Agentic Systems, Benchmarking, Model Evaluation, Data-centric AI.
Frameworks	PyTorch, Hugging Face Transformers, Accelerate, PEFT, DeepSpeed, Axolotl, LangChain, FAISS, AWS SageMaker, EC2.
Models	GPT-style Models, LLaMA, DeBERTa, ViT, Encoder-Decoder Models, VLMs.
Programming	Python, C++, SQL, Bash.
Languages	Vietnamese Native, English IELTS 7.0.

LEADERSHIP & COMMUNITY

- **Top 10 AI Talents**, AI Awards @ AI4VN 2025, hosted by VnExpress.
- **Invited Speaker**, AgentCon 2025 at SUTD and Google I/O Extended HCMC 2025.
- **AI Facilitator**, Build with AI Ho Chi Minh City 2024.
- **Chapter Lead**, Google Developer Student Clubs, 2022–2023.
- **Invited Speaker**, GDSC Summit 2022.
- **Community Contributor**, Google Developer Student Clubs, 2019–2023.

EDUCATION

VNU-Ho Chi Minh City University of Technology **2019 – 2023**
B.Sc. in Computer Science

Long An High School for the Gifted **2016 – 2019**
Specialization in Mathematics; Provincial Olympiad Scholar